WORKED SOLUTION

STEP II 1992, Question 7

Pure • Repeated Roots of Cubics

STEP II 1992, Question 7

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/92-s2-q7

The key idea

Comparing coefficients of $(x-a)(x-b)(x-c)$ with the cubic gives p, q and r as symmetric functions of the roots. In (i), a repeated root together with "roots sum to zero" pins the roots down to the shape $\alpha, \alpha, -2\alpha$, and everything follows from that. For (ii), the expression in the question looks like the answer to (i) with q and r replaced by more complicated things. That suggests shifting x to kill the x^2 term and then applying (i) to the new cubic.

Solution

Since a, b, c are the roots and the leading coefficient is 1,

$$x^3 - px^2 + qx - r = (x-a)(x-b)(x-c) = x^3 - (a+b+c)x^2 + (ab+bc+ca)x - abc$$

for all x. Comparing coefficients,

$$p = a+b+c, \qquad q = ab+bc+ca, \qquad r = abc.$$

Part (i)

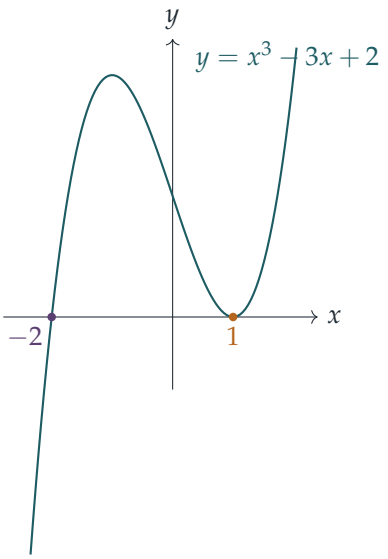
Label the roots so that the repeated one is $a = b = \alpha$. Then $p = 0$ gives $c = -(a+b) = -2\alpha$, so the roots are $\alpha, \alpha, -2\alpha$. Hence

$$q = \alpha^2 + \alpha(-2\alpha) + (-2\alpha)\alpha = -3\alpha^2,$$
$$r = \alpha \cdot \alpha \cdot (-2\alpha) = -2\alpha^3.$$

Therefore

$$4q^3 + 27r^2 = 4(-27\alpha^6) + 27(4\alpha^6) = -108\alpha^6 + 108\alpha^6 = 0,$$

as required. Nothing here needs α to be real, and the case of a triple root ($\alpha = 0$, all coefficients zero) is included.



For example, $x^3 - 3x + 2 = (x-1)^2(x+2)$ has $\alpha = 1$, $q = -3$, $r = -2$, and $4(-27) + 27 \cdot 4 = 0$. The double root shows up as the graph touching the axis.

Part (ii)

Now p need not be zero. The roots of the original cubic sum to p, so their mean is $p/3$. Shifting by the mean, put

$$x = y + \tfrac{p}{3}.$$

Then $y = x - \frac{p}{3}$, so the roots of the new cubic in y are $a - \frac{p}{3}$, $b - \frac{p}{3}$, $c - \frac{p}{3}$. Their sum is $p - p = 0$, and if two of a, b, c are equal then two of these are equal too.

Ask yourself What exactly is the cubic in y?

Expanding each term,

$$\begin{aligned} \left(y + \tfrac{p}{3}\right)^3 &= y^3 + py^2 + \tfrac{p^2}{3}y + \tfrac{p^3}{27}, \\ -p\left(y + \tfrac{p}{3}\right)^2 &= -py^2 - \tfrac{2p^2}{3}y - \tfrac{p^3}{9}, \\ q\left(y + \tfrac{p}{3}\right) &= qy + \tfrac{pq}{3}, \end{aligned}$$

and adding these together with $-r$, the y^2 terms cancel (that is the point of the shift) and

$$x^3 - px^2 + qx - r = y^3 + \left(q - \frac{p^2}{3}\right)y - \left(\frac{2p^3}{27} - \frac{pq}{3} + r\right),$$

using $\frac{p^3}{27} - \frac{p^3}{9} = -\frac{2p^3}{27}$. This cubic in y has the form $y^3 - Py^2 + Qy - R$ with

$$P = 0, \qquad Q = q - \frac{p^2}{3}, \qquad R = \frac{2p^3}{27} - \frac{pq}{3} + r.$$

It has zero y^2 coefficient and two equal roots, so part (i) applies to it and gives $4Q^3 + 27R^2 = 0$, that is,

$$4\left(q - \frac{p^2}{3}\right)^3 + 27\left(\frac{2p^3}{27} - \frac{pq}{3} + r\right)^2 = 0,$$

which is the required result.

Checks

- With $p = 0$ the result of (ii) reduces to $4q^3 + 27r^2 = 0$, which is part (i).
- For $(x-1)^2(x-2) = x^3 - 4x^2 + 5x - 2$ we have $p = 4$, $q = 5$, $r = 2$, so $Q = -\frac{1}{3}$ and $R = \frac{128}{27} - \frac{20}{3} + 2 = \frac{2}{27}$. Then $4Q^3 + 27R^2 = -\frac{4}{27} + \frac{4}{27} = 0$.
- We also tested both identities in exact rational arithmetic on hundreds of random cubics with a repeated root, and checked the shift identity at random points.

Common slips

- Sign errors in Vieta's formulae: with the signs $-p$, $+q$, $-r$ chosen in this question, all three of p, q, r come out positive in terms of the roots.
- Shifting by $p/3$ in the wrong direction. The roots of the new cubic are $a - \frac{p}{3}$ and so on, which is what makes their sum zero.
- Expanding everything in (ii) in terms of a, a, c by brute force. It works, but it is long and error-prone; the shift reuses (i) for free.

Beyond STEP

For a general cubic the left-hand side of (ii) equals $-(a-b)^2(b-c)^2(c-a)^2$, the negative of the *discriminant*. It vanishes exactly when two roots coincide, and its sign tells you whether a real cubic has three real roots or only one. The shift $x = y + \frac{p}{3}$ is also the first step of Cardano's formula for solving cubics. Symmetric functions of roots, discriminants and the question of which equations can be solved by radicals lead into Galois theory; Allenby's *Rings, Fields and Groups* is a good place to start.

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